Transcription and Translation

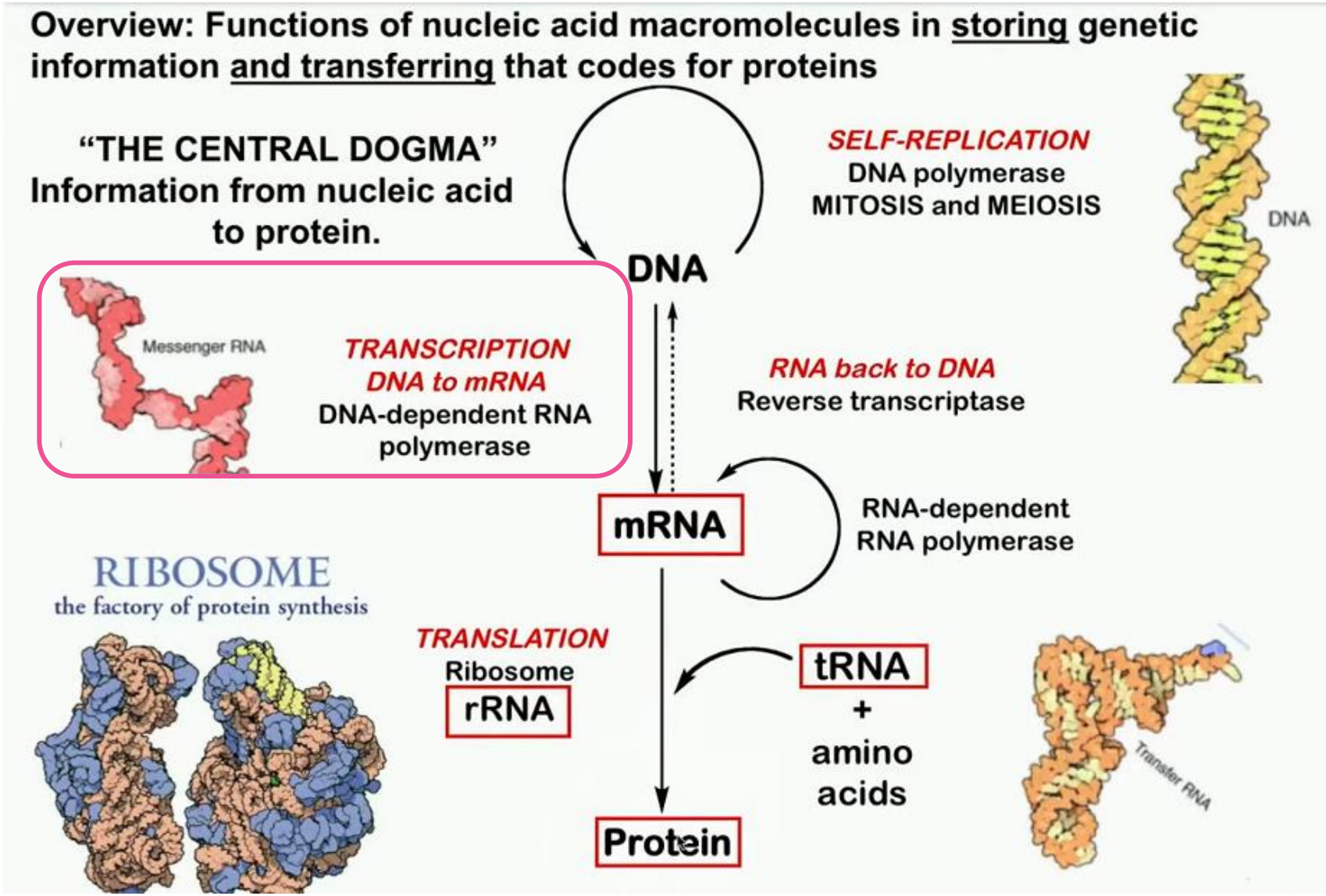
From DNA sequence to functional protein

Lecture 17

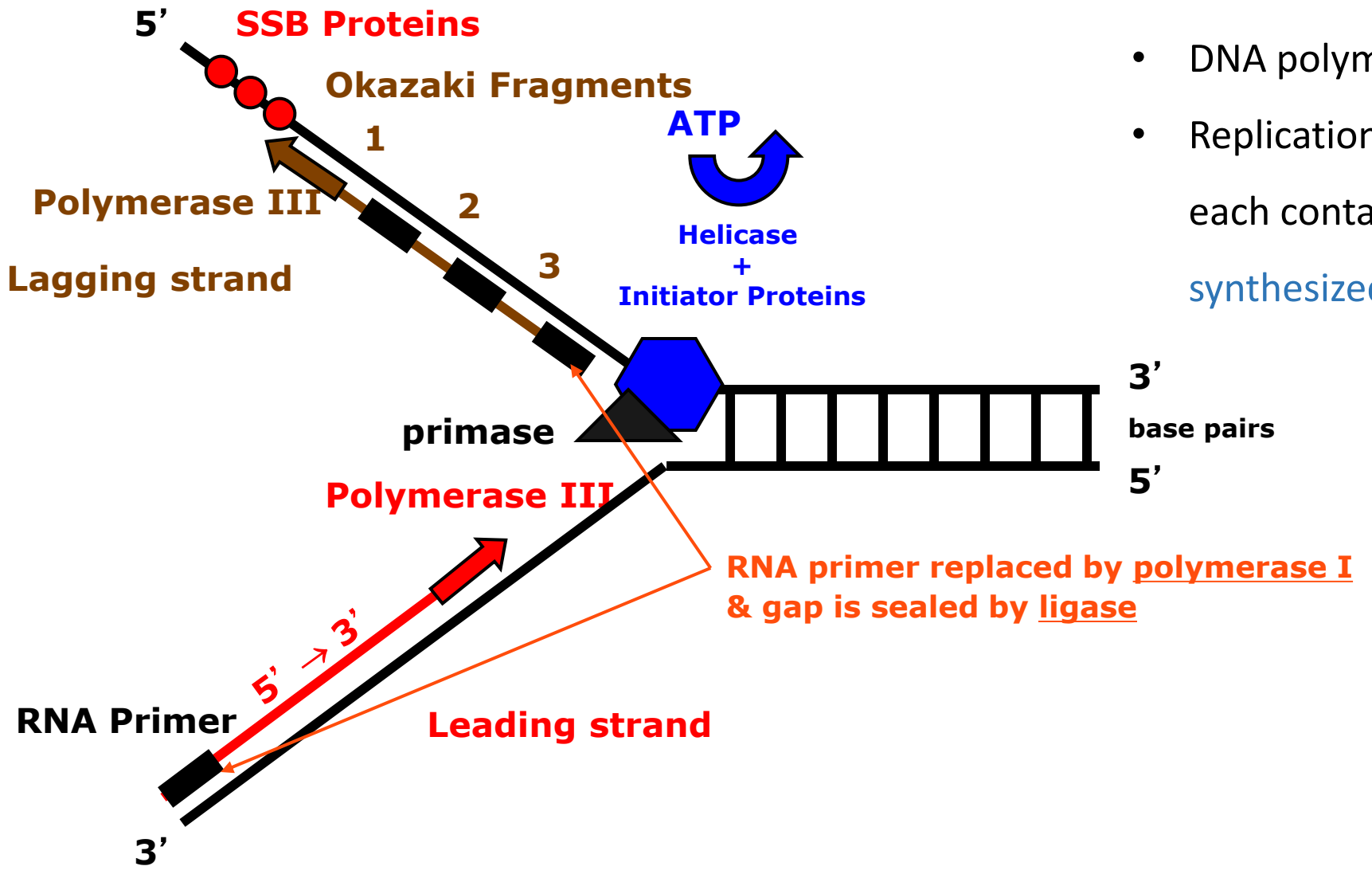
Prof. Saurabh Raj
Kusuma School of Biological Sciences
IIT Delhi

Contact: *sraj@iitdabudhabi.ac.ae*

The molecules of life

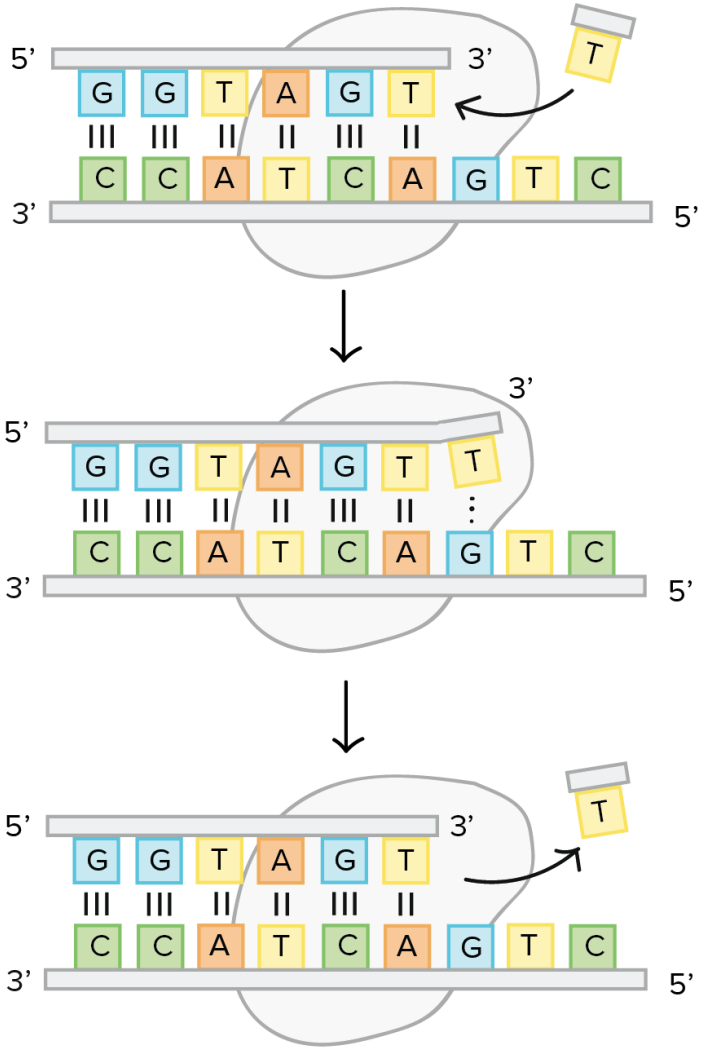


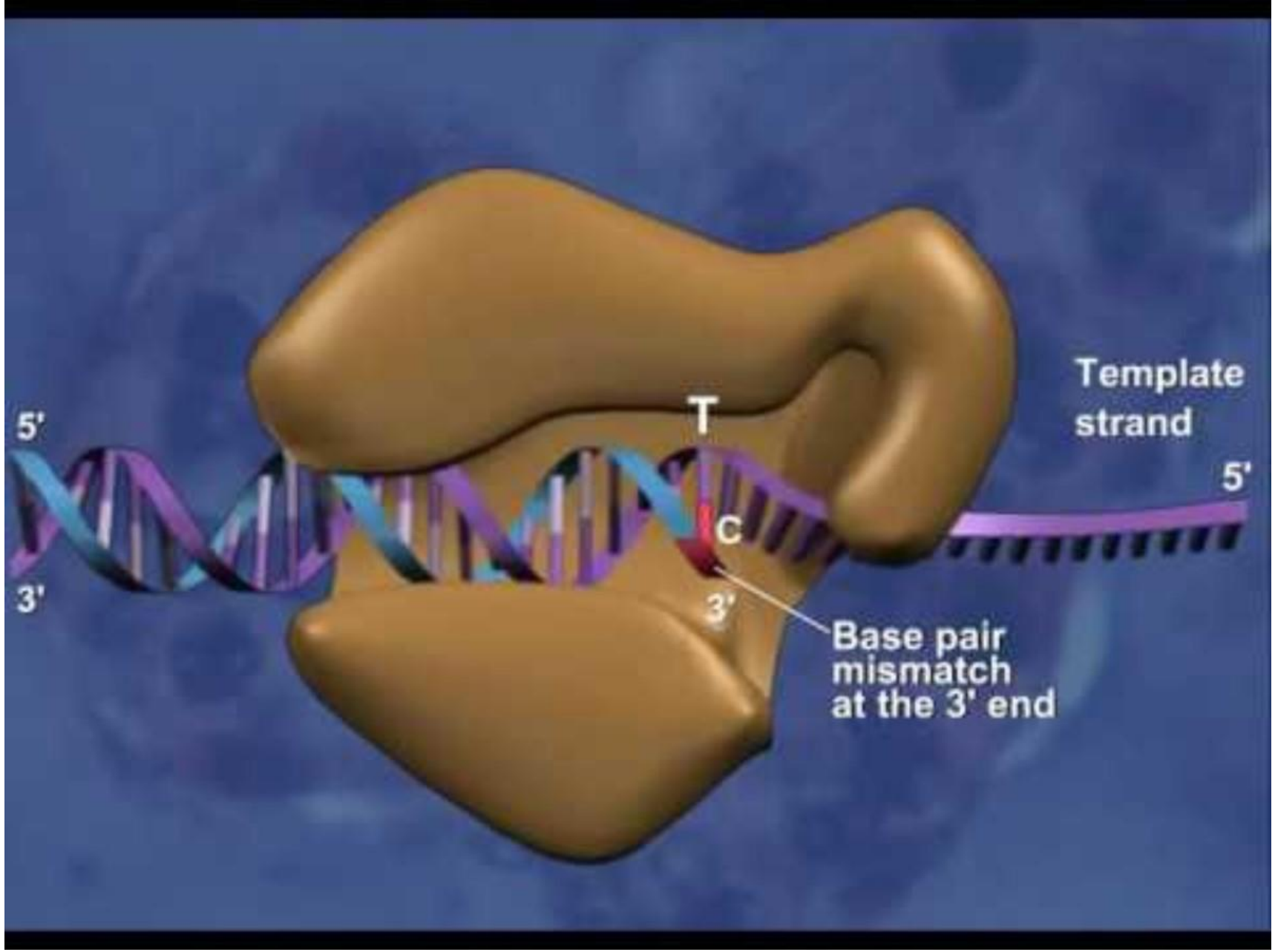
DNA Replication — Quick Recap



- DNA replication is bidirectional
- There is leading and lagging strand
- DNA polymerase synthesizes DNA 5′ → 3′
- Replication produces two DNA molecules, each containing one parental and one newly synthesized strand

- Error rate of DNA polymerase is roughly 1 in 10,000-100,000.
- DNA polymerase can do proofreading – 3’→5’ exonuclease activity.
- Proofreading decreases the error rate significantly – from 1 in 10,000 to 1 in 10^7-10^8



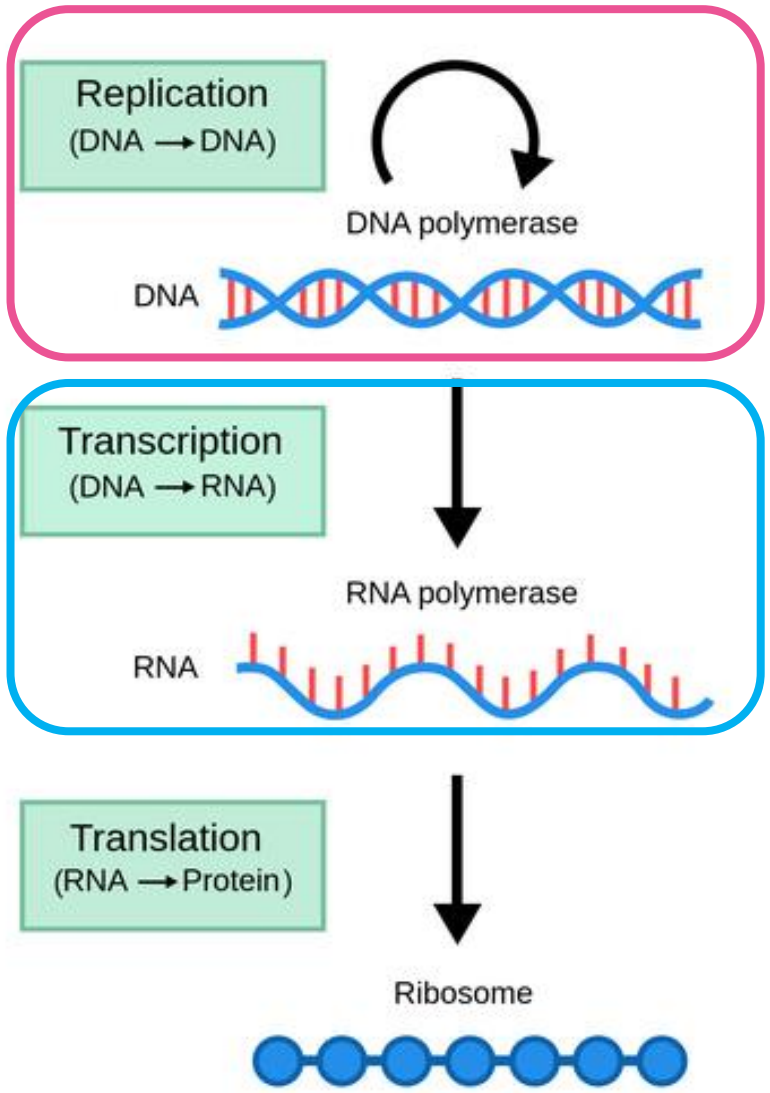


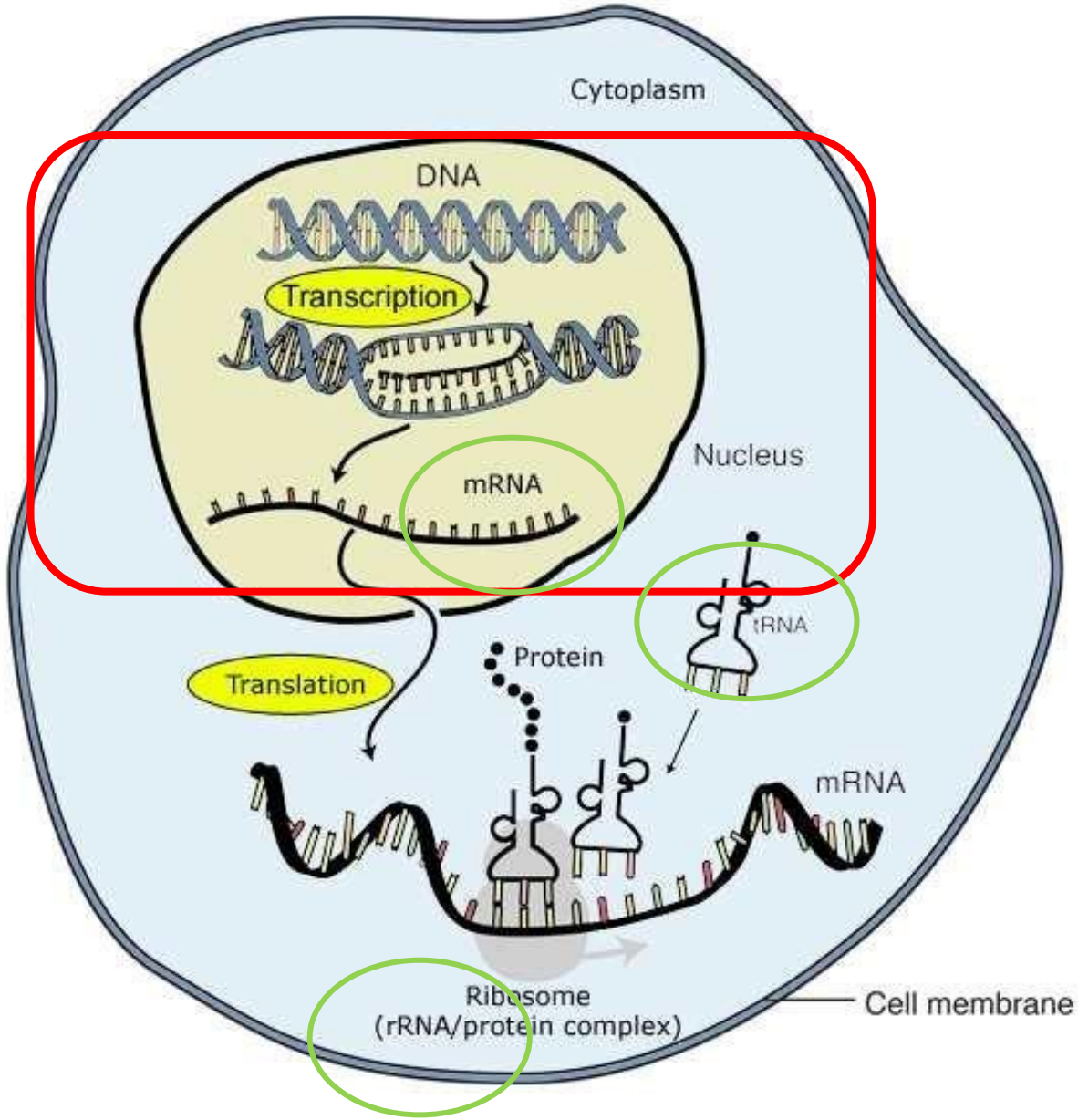
https://www.youtube.com/watch?v=03H3t6jD6yk



Transcription

Transcription DNA → RNA





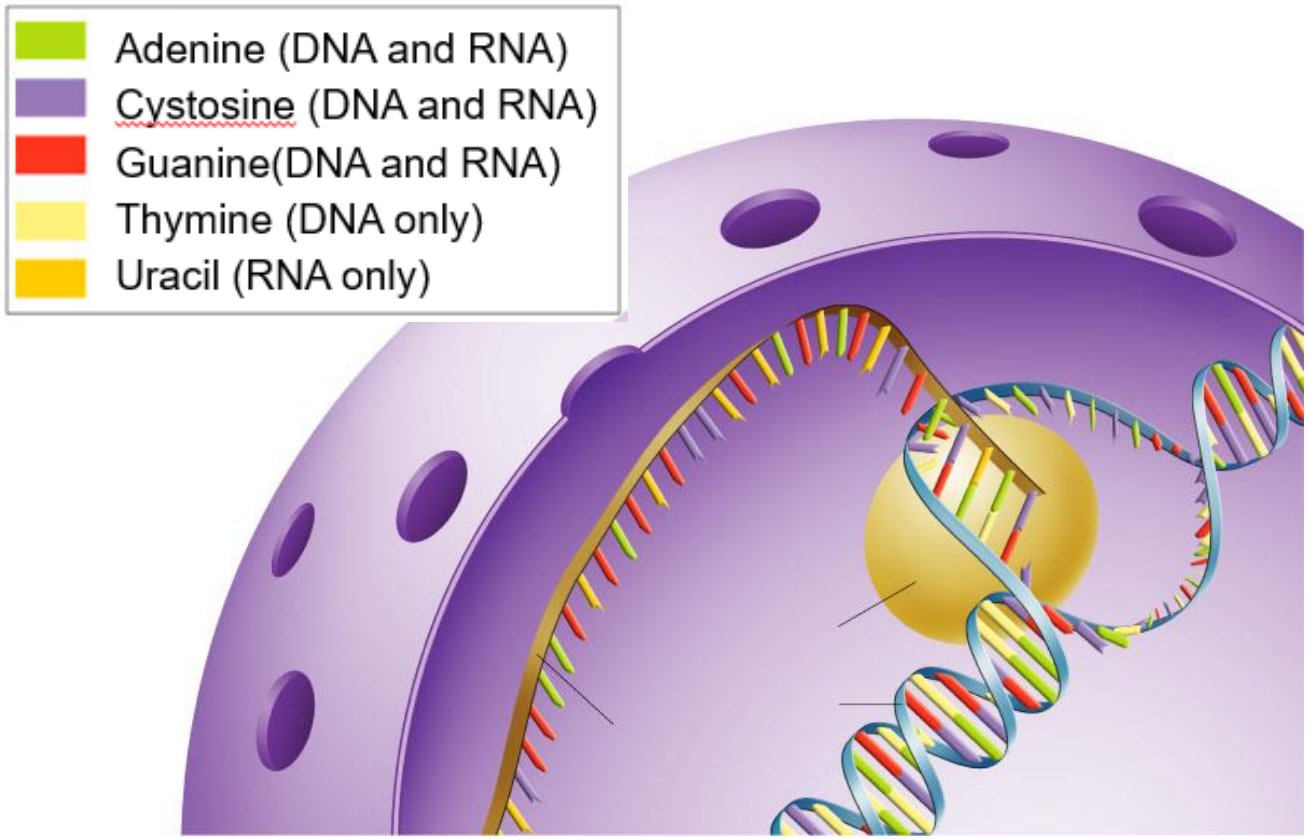
DNA and RNA

DNA: A, T, G and C

- **Sugar**: Deoxyribose
- **Stores Genetic information as dsDNA**
- **Mostly found in Nucleus and mitochondria**

RNA: A, U, G and C

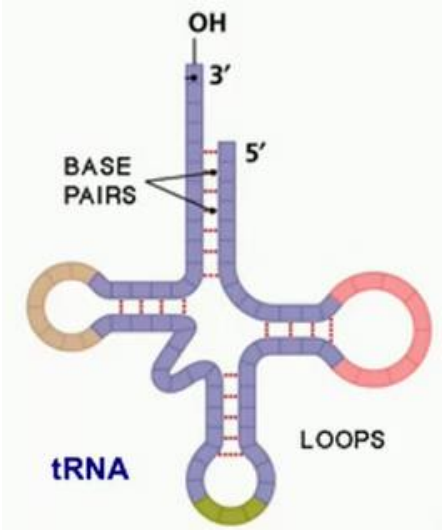
- **Sugar: Ribose**
- **Found through out the cells**
- **RNA involved in protein synthesis are:**
  - **Messenger RNA (mRNA)**
  - **transfer RNA (tRNA)**
  - **ribosomal RNA (rRNA)**



Why RNA?

RNA is more than a messenger.

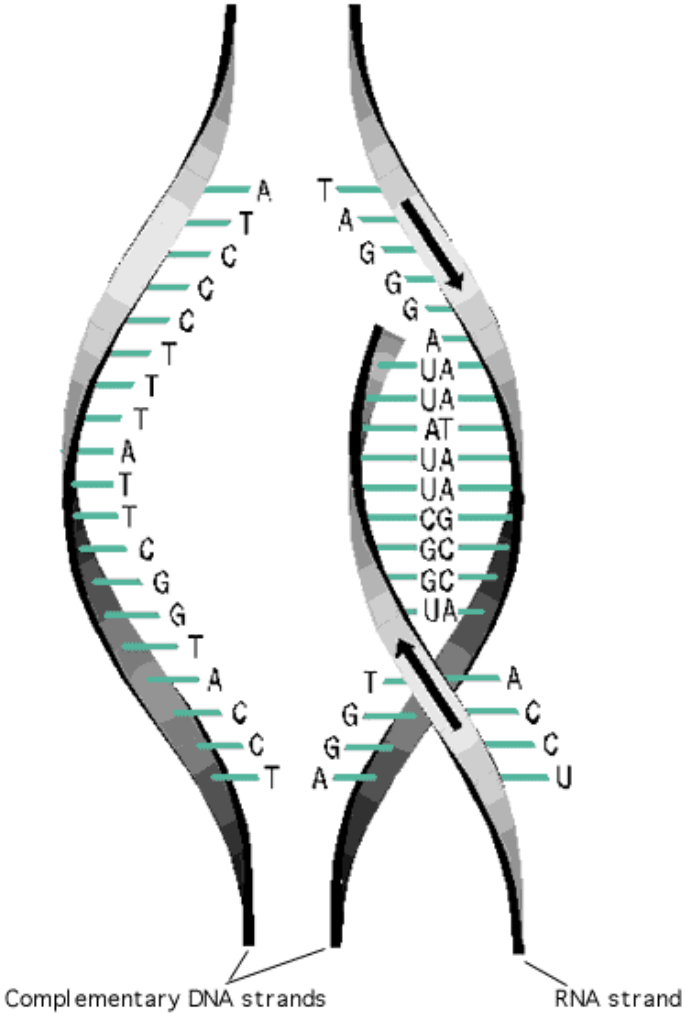
- **mRNA** — carries genetic information from DNA
- **tRNA** — connects codons with amino acids
- **rRNA** — forms the core of the ribosome and participates in peptide-bond formation
- Other RNAs regulate gene expression and cellular processes



DNA stores information; RNA helps use and regulate that information.

Transcription

Transcription = synthesis of RNA using DNA as a template



During transcription:

- RNA nucleotides base-pair with the DNA template
- **A** pairs with **U**
- **T** pairs with **A**
- **G** pairs with **C**
- **C** pairs with **G**

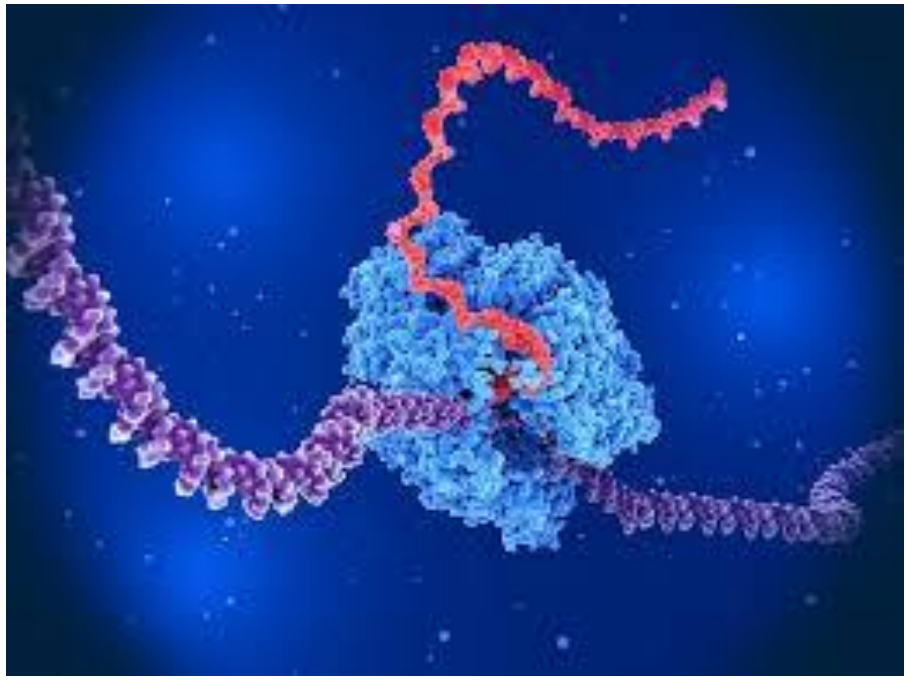
The initial RNA molecule produced is called the **primary transcript**.

TRANSCRIPTION

ACGATACCCTGACGAGCGTTAGCTATCG

UGC UAU GGG ACU

Major players in transcription: RNA Polymerase



Transcription of DNA Inside the Nucleus

1. **DNA unzips.**
2. **mRNA bases attach.**
3. **mRNA goes to ribosome.**
4. **DNA zips up again.**

- RNA polymerase- complex of enzymes with 2 functions:
  - Unwind DNA sequence
  - Produce primary transcript by stringing together the chain of RNA nucleotides

Transcription – from Gene to RNA

Not all DNA is transcribed.

- A **gene/transcription unit** contains the DNA information used to produce an RNA molecule
- RNA polymerase transcribes the selected region
- The resulting RNA may be **processed**
- Some RNAs function directly; others, such as mRNA, are translated into protein

In eukaryotic cells

DNA → primary RNA transcript → processed RNA → cytoplasm → translation

In prokaryotes, transcription and translation can occur in the same cellular compartment.

Transcription

- Carried out by **RNA polymerase**
    - RNA polymerase synthesizes RNA from 5'→ 3'
    - Does not need a primer
    - Includes helicase activity
- Only one of the two DNA strands are transcribed
- Only part of DNA is transcribed
- Steps involve:
    - **INITIATION**: RNA polymerase binds to a special region (DNA sequence) called the promoter to guide which part of DNA to use as the template.
        - RNA polymerase is recruited to a promoter
        - DNA is locally unwound
        - RNA synthesis begins
    - **ELONGATION**: extension from 5' to 3'
        - RNA polymerase moves along the DNA template
        - The RNA strand is extended 5' → 3'
    - **TERMINATION:** RNA polymerase encounters a terminator sequence on the DNA.
        - RNA polymerase reaches a termination signal
        - The newly synthesized RNA is released

Transcription

Coding strand

5′ – ACG ATA CCC TGA CGA GCG TTA GCT ATC G – 3′

Template strand

3′ – TGC TAT GGG ACT GCT CGC AAT CGA TAG C – 5′

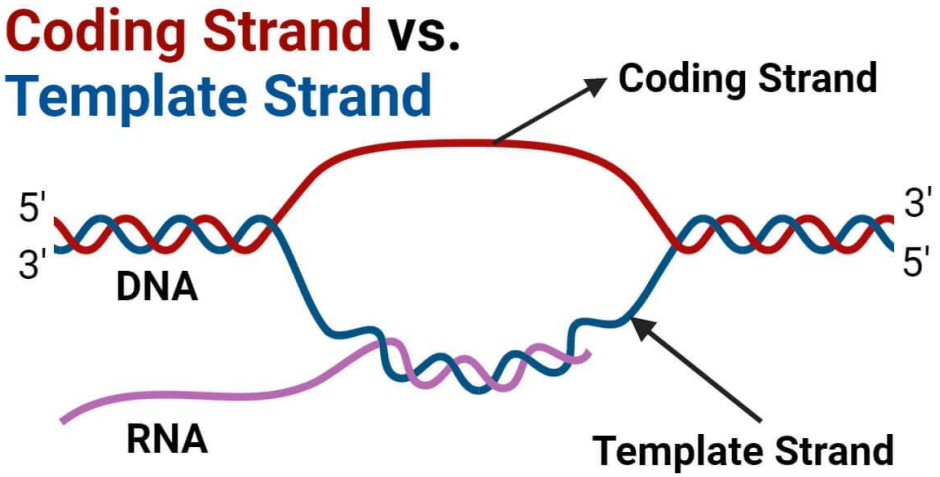
↓

RNA polymerase

↓

mRNA

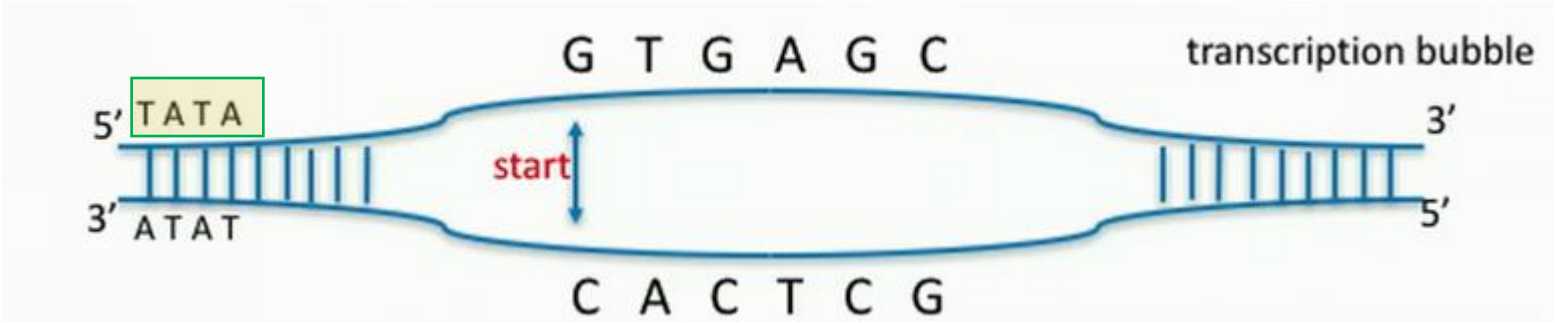
5′ – ACG AUA CCC UGA CGA GCG UUA GCU AUC G – 3′



The mRNA sequence is complementary to the template strand and almost identical to the coding strand, except U replaces T.

⚠ For a given gene, **one DNA strand serves as the template**.
Different genes can use different DNA strands as templates.

Transcription Initiation

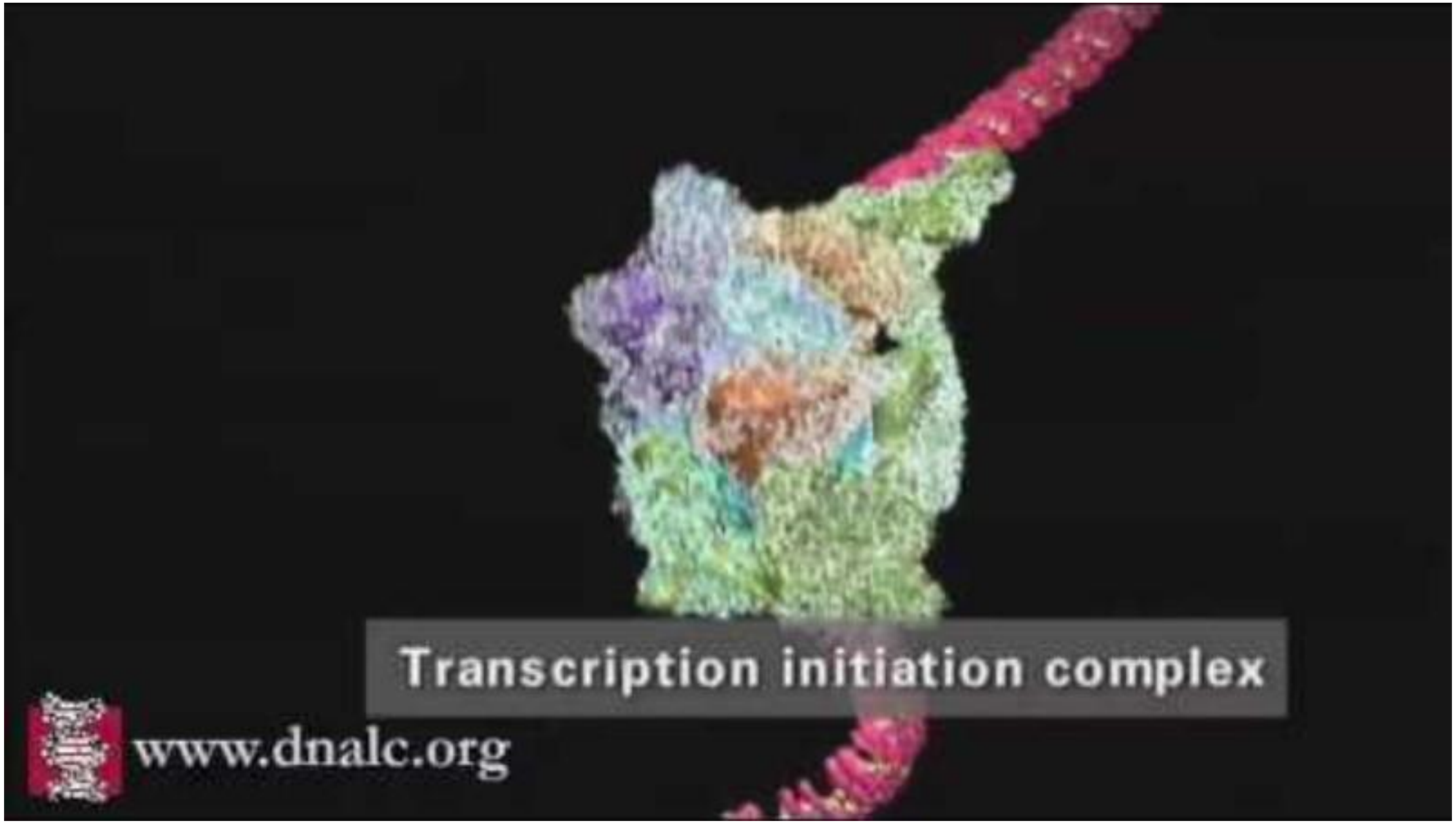


Which strand is used as the template for mRNA synthesis?

a) Top Strand

b) Bottom Strand

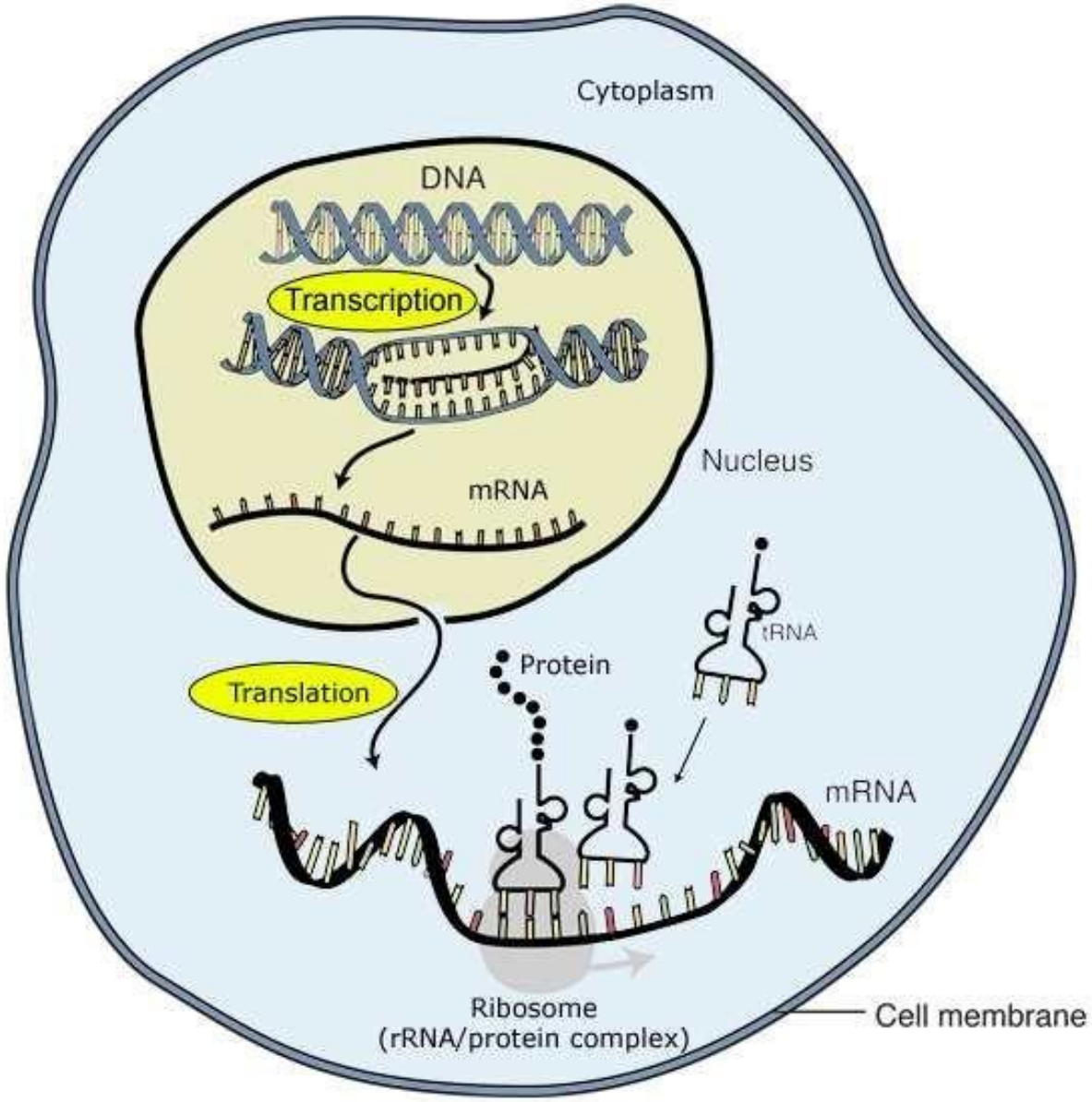
ALWAYS REMEMBER: You read 3’ to 5’ and you make 5’ to 3’!


Transcription initiation complex
www.dnalc.org

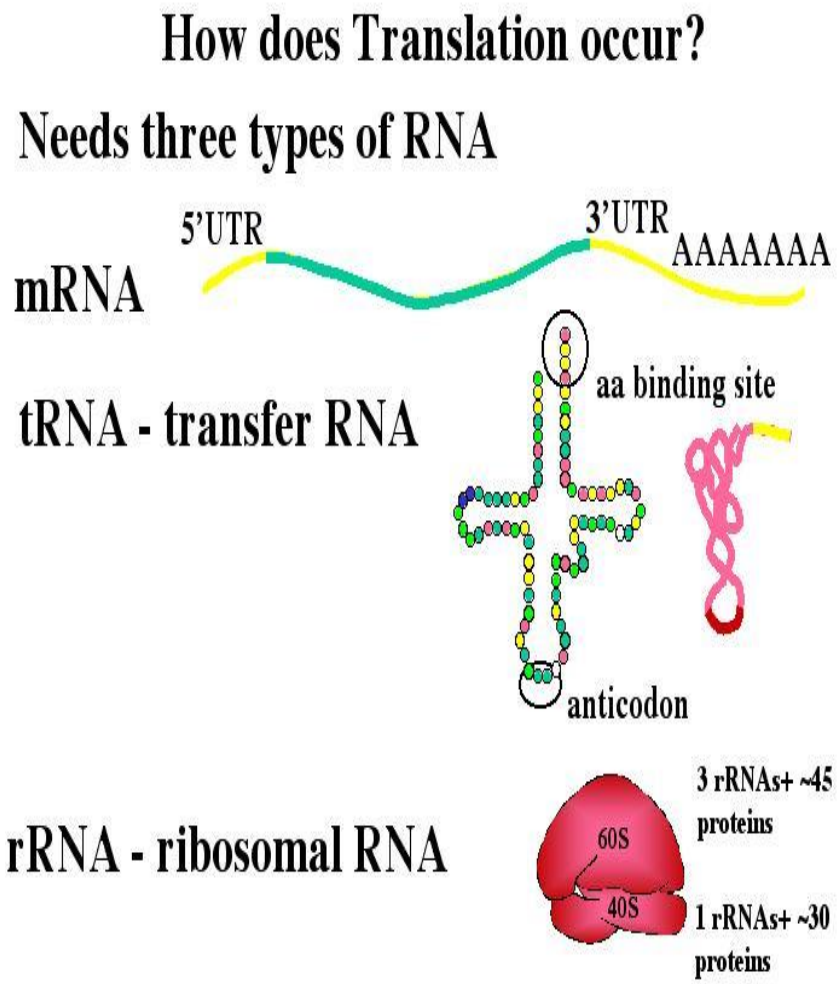


TRANSLATION

RNA involved in protein synthesis



RNA involved in protein synthesis



***DNA** = STORING GENETIC INFORMATION*

RNA Molecules Involved in Protein Synthesis

mRNA — Messenger RNA

- Carries the genetic information required to make a protein
- Contains the sequence of **codons**

tRNA — Transfer RNA

- Carries amino acids to the ribosome
- Uses an **anticodon** to recognize the appropriate mRNA codon

rRNA — Ribosomal RNA

- Forms the structural and catalytic core of the ribosome
- Participates directly in protein synthesis

Remember

mRNA = message
tRNA = transporter (or decoder)
rRNA = ribosome

TRANSLATION

- Involves the transfer of biological information from the 4 nucleotide code of nucleic acids [A,C,G,T (U)] into the 20 amino acid code of proteins

  - 1 base = 1 amino acid

    $4^1 = 4$

  - 2 bases = 1 amino acid

    $4^2 = 16$

  - 3 bases = 1 amino acid

    $4^3 = 64$

TRANSLATION

- Involves the transfer of biological information from the 4 nucleotide code of nucleic acids [A,C,G,T (U)] into the 20 amino acid code of proteins

- Each 3 nucleotide coding unit that specifies an amino acid is called a codon. A codon is a group of three consecutive nucleotides required to specify a single amino acid

The Genetic Code

First position	Second position: U		Second position: C		Second position: A		Second position: G		Third position
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
	UUC	Phe	UCC	Ser	UAC	Tyr	UGC	Cys	C
	UUA	Leu	UCA	Ser	UAA	*Stop*	UGA	*Stop*	A
	UUG	Leu	UCG	Ser	UAG	*Stop*	UGG	Trp	G
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
	CUC	Leu	CCC	Pro	CAC	His	CGC	Arg	C
	CUA	Leu	CCA	Pro	CAA	Gln	CGA	Arg	A
	CUG	Leu	CCG	Pro	CAG	Gln	CGG	Arg	G
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U
	AUC	Ile	ACC	Thr	AAC	Asn	AGC	Ser	C
	AUA	Ile	ACA	Thr	AAA	Lys	AGA	Arg	A
	AUG	Met	ACG	Thr	AAG	Lys	AGG	Arg	G
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U
	GUC	Val	GCC	Ala	GAC	Asp	GGC	Gly	C
	GUA	Val	GCA	Ala	GAA	Glu	GGA	Gly	A
	GUG	Val	GCG	Ala	GAG	Glu	GGG	Gly	G

ACGATACCCTGACGAGCGTTAGCTATCG

UGC UAU GGG ACUG

First position	Second position U		Second position C		Second position A		Second position G		Third position
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
	UUC	Phe	UCC	Ser	UAC	Tyr	UGC	Cys	C
	UUA	Leu	UCA	Ser	UAA	*Stop*	UGA	*Stop*	A
	UUG	Leu	UCG	Ser	UAG	*Stop*	UGG	Trp	G
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
	CUC	Leu	CCC	Pro	CAC	His	CGC	Arg	C
	CUA	Leu	CCA	Pro	CAA	Gln	CGA	Arg	A
	CUG	Leu	CCG	Pro	CAG	Gln	CGG	Arg	G
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U
	AUC	Ile	ACC	Thr	AAC	Asn	AGC	Ser	C
	AUA	Ile	ACA	Thr	AAA	Lys	AGA	Arg	A
	AUG	Met	ACG	Thr	AAG	Lys	AGG	Arg	G
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U
	GUC	Val	GCC	Ala	GAC	Asp	GGC	Gly	C
	GUA	Val	GCA	Ala	GAA	Glu	GGA	Gly	A
	GUG	Val	GCG	Ala	GAG	Glu	GGG	Gly	G

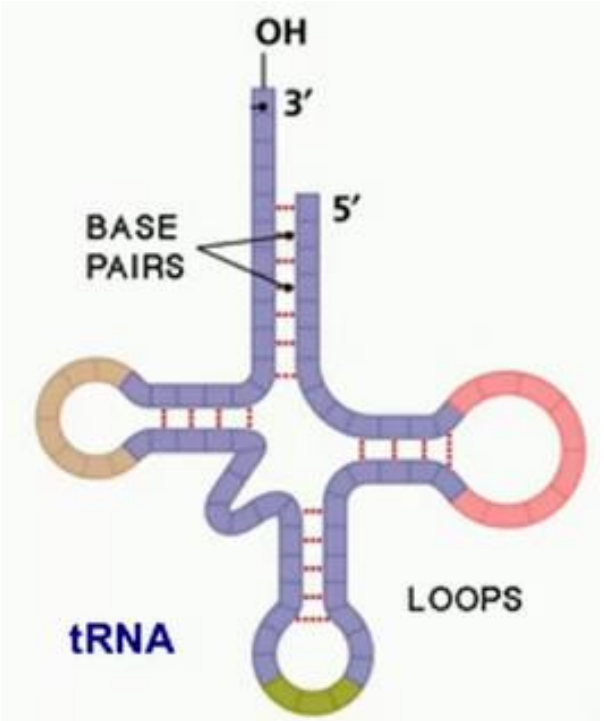
TRANSLATION Summary

- Ribosomes translates the information encoded in mRNA to form protein
- The nucleotide sequence in the mRNA is decoded with tRNAs
- Main Players :
  - mRNA,
  - tRNAs and
  - amino acid, Ribosomes
- Key steps:
  - Initiation (START),
  - Elongation and
  - Termination (STOP)

TRANSLATION

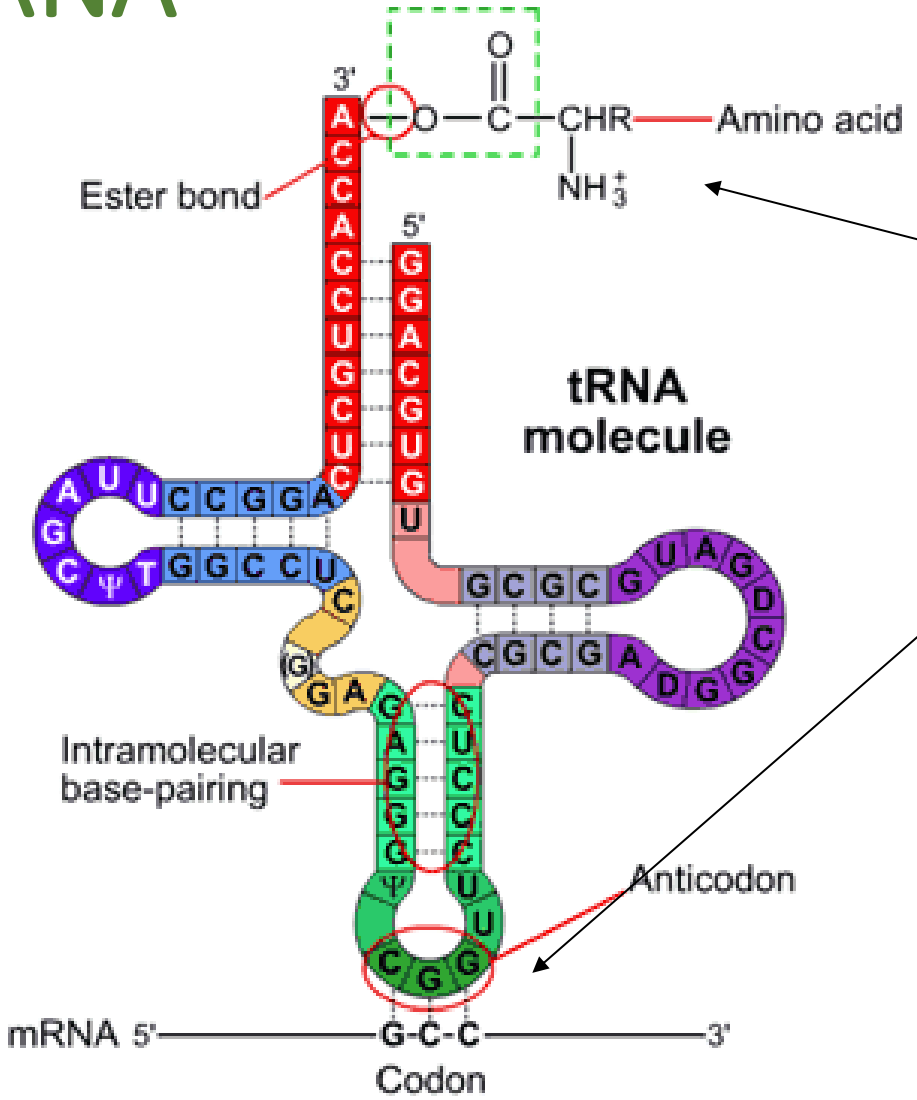
- mRNA, a ribosome, and the first tRNA molecule come together.

- The tRNA containing MET will recognize the ***start*** signal and start the TRANSLATION process.



anticodon

tRNA



- Transfer RNA
- Bound to one amino acid on one end
- Anticodon on the other end complements mRNA codon
- Amino acids must be in the correct order for the protein to function correctly
- tRNA lines up amino acids using mRNA code

TRANSLATION cont.

TERMINATION

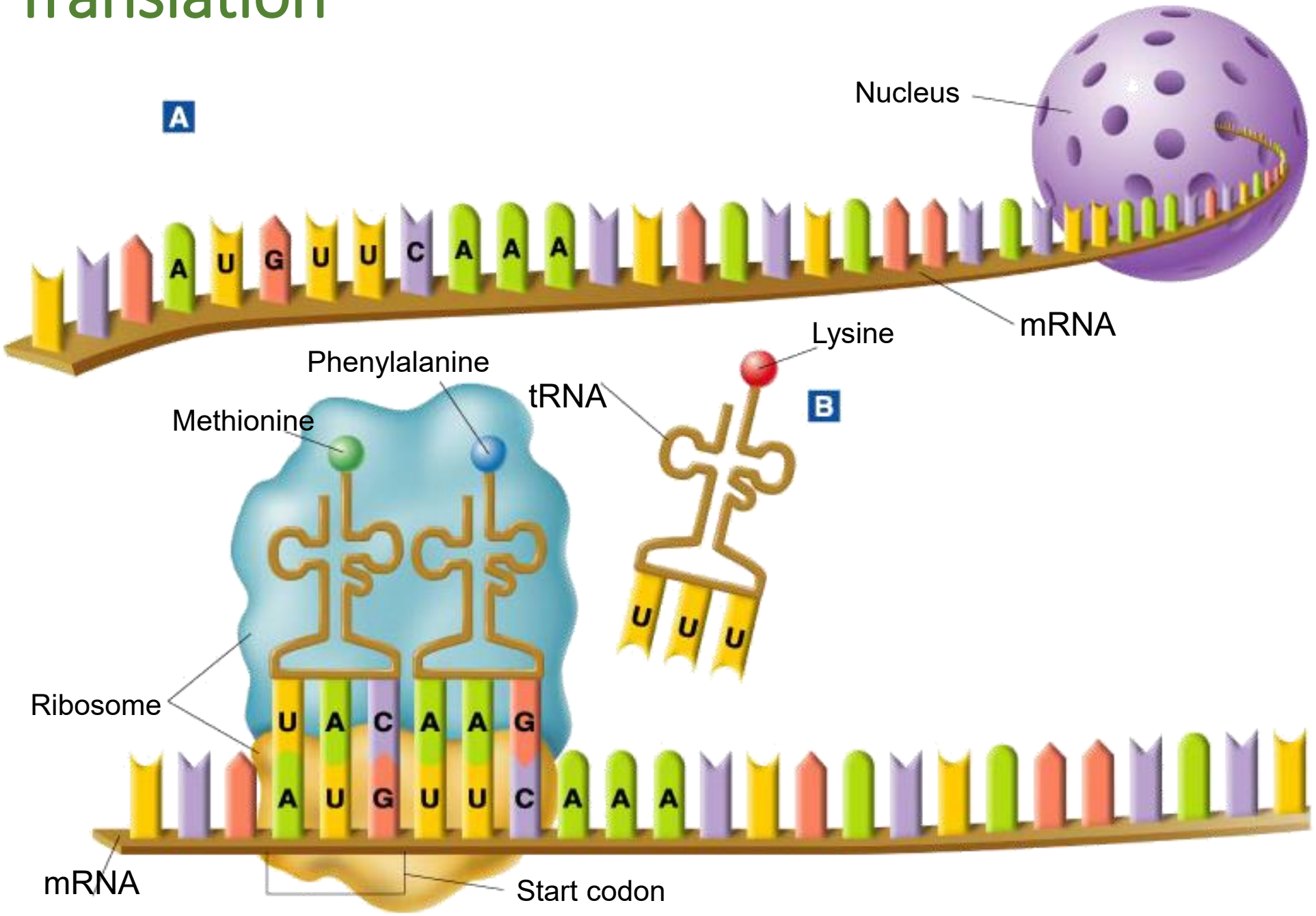
- A "**stop**" codon (UAA, UGA, or UAG) signals the end of the process.

- An enzyme called the ***releasing factor*** frees the amino acid chain

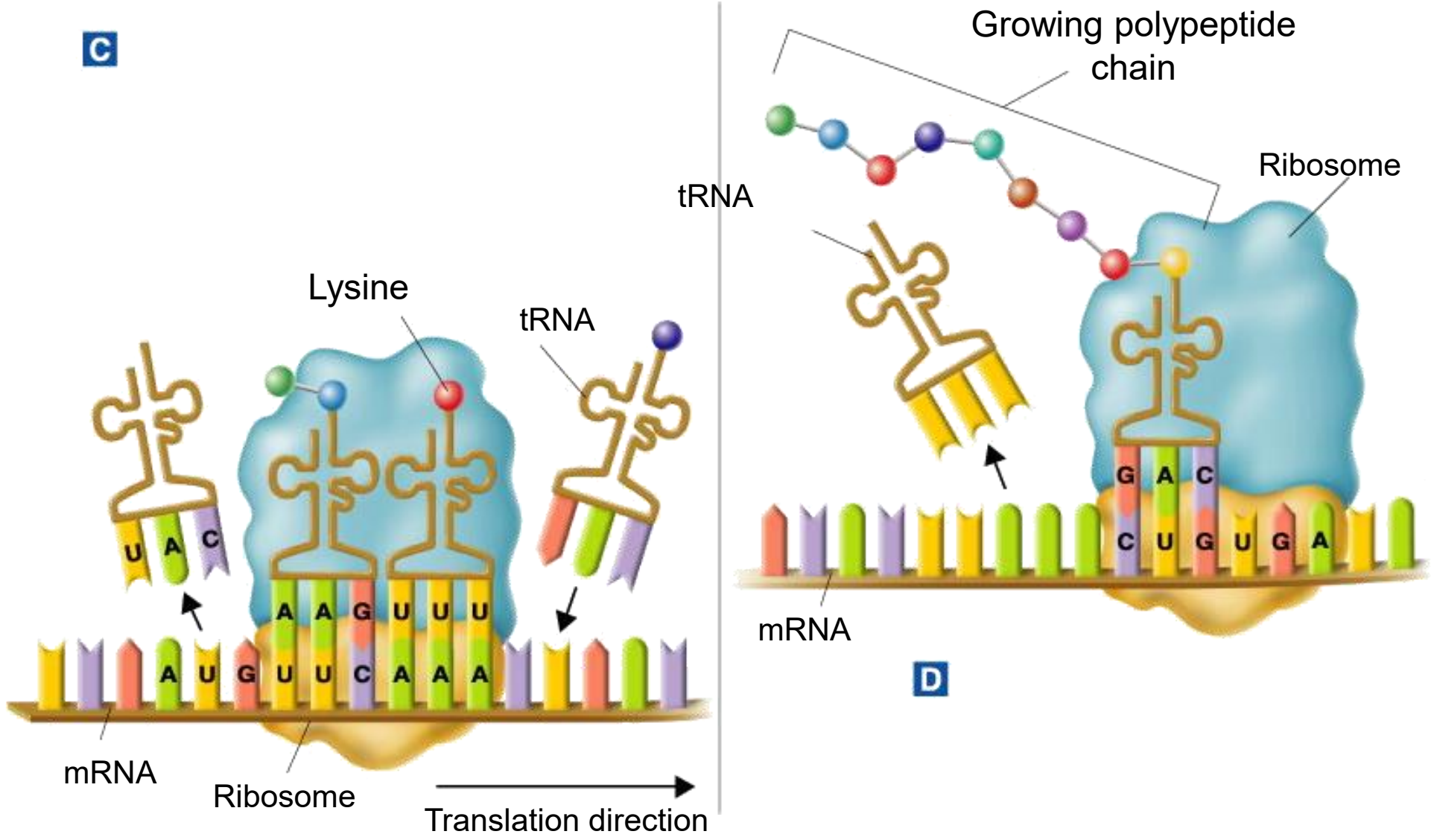
- The mRNA molecule is released from the ribosome complex which then falls apart

- All mRNA are eventually degraded when the protein no longer needs to be made.

Translation



Translation (continued)



TRANSLATION

Translation process occurs in ribosomes where mRNA is converted into proteins

